

CRITICAL ARCHITECTURAL BOUNDARY: The production application remains **Next.js 16 (App Router + Turbopack + React 19 + SQLite/Prisma)**. Module 8 requirements are demonstrated via an isolated, parallel Vite workspace (module-8-vite/). Locked Module 6 RSC Architecture: **S = 26, C = 11, Total = 37 (70.27% RSC)**. Verified 100% intact.

1. TOOLCHAIN & ARCHITECTURE MIGRATION MATRIX

Technical Area	Production Project (nextjs-app/)	Module 8 Workspace (module-8-vite/)	Status
Framework	Next.js 16.3.5 (Hybrid RSC + Client)	React 19.2.8 (Client SPA Architecture)	PARALLEL
Build Engine	Turbopack Engine (Rust)	Vite 6.4.3 (Esbuild Go pre-bundler + Rollup)	PASS
Dev Server	Turbopack incremental route compile	Native ECMAScript Modules (ESM) on-demand	PASS
Lint & Format	ESLint 9 (Flat Config, ~45s run)	Biome 1.8.3 (Rust-based, 6 ms run)	PASS
TypeScript	TypeScript 5.7 (strict: true)	Strict + noUncheckedIndexedAccess	PASS
Path Alias	@/* mapped to root	@/* mapped to module-8-vite/src/*	PASS
Code Splitting	Automatic Next.js route chunks	Rollup manualChunks (vendor separation)	VERIFIED

2. OPTIMIZED VITE CONFIGURATION (VITE.CONFIG.TS)

- **React Plugin:** @vitejs/plugin-react enables Fast Refresh and JSX AST transformation.
- **Path Aliasing:** resolve.alias: { '@': path.resolve(__dirname, './src') } verified across all source imports.
- **Dev Server:** port: 3000, strictPort: true, host: true for predictable local networking.
- **Production Target:** target: 'esnext', outDir: 'dist', minify: 'esbuild', sourcemap: false.
- **Vendor Splitting:** rollupOptions.output.manualChunks: { vendor: ['react', 'react-dom'] } guarantees independent long-term caching for core React dependencies.

3. RUST TOOLCHAIN: BIOME INTEGRATION (BIOME.JSON)

Biome Configuration Feature	Specification in biome.json	Execution Outcome
Import Organization	organizeImports: { enabled: true }	Auto-sorted in 260ms during init
Linter & Correctness Rules	recommended: true, noUnusedVariables: 'error'	0 errors across all 9 files in 6ms
Code Formatting	indentStyle: 'space', indentWidth: 2, lineWidth: 100	Fully standardized styling
Performance Contrast vs ESLint	Rust multi-threaded parser vs JS Node.js AST runtime	> 4,000× speedup (6ms vs 45s)

4. EMPIRICAL BUILD & RUNTIME BENCHMARKS (OBSERVED MEASUREMENTS)

Benchmark Metric	Measured Value	Target / Ref	Empirical Analysis & Execution Environment
Vite Cold Start	423 ms	< 500 ms	Vite 6.4.3 instant dev socket ready (4.58× faster than Next.js 1,939ms).
HMR Response Latency	Sub-second (< 500ms)	Sub-second	Edits in HmrTester.tsx broadcasted & updated in the same clock second.
Vite Bundling Time	1.54 s – 3.01 s	< 5.0 s	33 modules transformed, minified, gzip computed via Esbuild.
Total Build Pipeline	9.37 seconds	< 15.0 s	Combines strict TypeScript checking (tsc -b 7.8s) + Vite bundler (1.54s).
Production Next.js Build	~60 seconds	< 90 s	Full Turbopack SSG generation for 18 routes (retains 100% integrity).

5. PHYSICAL CODE SPLITTING & DIST OUTPUT VERIFICATION

Generated Chunk Artifact	Disk Size	Gzip Size	Chunk Separation Verification Role
dist/assets/vendor-BZPdts19.js	12.35 kB	4.34 kB	Core React 19 + ReactDOM isolated via manualChunks.
dist/assets/index-IBJguqNi.js	216.49 kB	67.79 kB	Application UI components, state management, analytics logic.
dist/assets/index-BW1u5F6K.css	0.36 kB	0.28 kB	Standalone CSS stylesheet extracted and minified.
dist/index.html	0.57 kB	0.36 kB	Clean HTML entry point referencing native ESM hashes.

6. STRICT TYPESCRIPT, STATIC QUALITY & PROTECTED INVARIANTS

Verification Area	Target Standard	Observed Status	Verification Command & Invariant Detail
Strict TypeScript	strict: true, noUncheckedIndexedAccess	PASS (0 errors)	npx tsc -b exits with code 0. Zero implicit any.
Biome Linter	Recommended rules, 0 errors	PASS (0 errors)	npx @biomejs/biome check ./src checked 9 files in 6ms.
SonarQube Quality	Static config integration	CONFIGURED	sonar-project.properties created; local scanner server offline.
CI/CD Pipeline	GitHub Actions workflow	PASS	.github/workflows/ci.yml configured for dual verification.
Module 6 RSC Invariant	S = 26, C = 11, T = 37 (70.27%)	PROTECTED	Server: 26, Client: 11 (26/37 = 70.27%). 0 regressions.
Next.js Production App	Build 18 routes successfully	PROTECTED	npm run build exits code 0. Turbopack SSR/SSG intact.